

5.16 Classwork: Logarithms Practice (no calculator)

1. [Maximum mark: 12]

(a) Calculate the value of each of the following logarithms:

(i) $\log_5 125$

(ii) $\log_{16} \frac{1}{4}$

(iii) $\log_3 \sqrt{81}$ [6]

(b) Let $u = \log_2 x$, where $x > 0$. Write down each of the following expressions in terms of u .

(i) $\log_2 x^3$

(ii) $\log_2(8x)$

(iii) $\log_4 x$ [6]

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2. [Maximum mark: 12]

(a) Express each in terms of $\log x$ and $\log y$.

(i) $\log(100x)$

(ii) $\log\left(\frac{x^2}{y}\right)$

(iii) $\log \sqrt{xy^3}$

[6]

(b) Write each of the following as a single logarithm.

(i) $\log_2 x + \log_2 y$

(ii) $2\log_7 a - \log_7 b$

(iii) $\frac{1}{2}\ln p + \ln q$

[6]

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